River Test

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Ferry travelling across a river(20 points)

1. (4 points) A ferry is crossing a river that is 18m wide at a velocity of $9\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $3\,\mathrm{ms}^{-2}$ down relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
2. (4 points) A ferry is crossing a river that is 168m wide at a velocity of $24\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $8\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
3. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $24\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $4\,\mathrm{ms}^{-2}$ down relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
4. (4 points) A ferry is crossing a river that is 8m wide at a velocity of $1\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $7\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
5. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $4\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $5\,\mathrm{ms}^{-2}$ down relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat

Ferry travelling straight across a river(15 points)

6. (3 points) A boat is crossing a river that is 9m wide with a velocity of $1\,\mathrm{ms}^{-2}$ relative to the water. However, the water is travelling at $1\,\mathrm{ms}^{-2}$ down relative to the ground. If the boat wants to travel in a straight line across, calculate:
   (a) The resultant velocity of the boat relative to the ground
   (b) The required angle the boat needs to travel at to cross the river directly across
   (c) The time taken to cross the river
7. (3 points) A boat is crossing a river that is 49m wide with a velocity of $7\,\mathrm{ms}^{-2}$ relative to the water. However, the water is travelling at $1\,\mathrm{ms}^{-2}$ up relative to the ground. If the boat wants to travel in a straight line across, calculate:
   (a) The resultant velocity of the boat relative to the ground
   (b) The required angle the boat needs to travel at to cross the river directly across
   (c) The time taken to cross the river
8. (3 points) A boat is crossing a river that is 160m wide with a velocity of $20\,\mathrm{ms}^{-2}$ relative to the water. However, the water is travelling at $1\,\mathrm{ms}^{-2}$ down relative to the ground. If the boat wants to travel in a straight line across, calculate:
   (a) The resultant velocity of the boat relative to the ground
   (b) The required angle the boat needs to travel at to cross the river directly across
   (c) The time taken to cross the river
9. (3 points) A boat is crossing a river that is 60m wide with a velocity of $15\,\mathrm{ms}^{-2}$ relative to the water. However, the water is travelling at $9\,\mathrm{ms}^{-2}$ down relative to the ground. If the boat wants to travel in a straight line across, calculate:
   (a) The resultant velocity of the boat relative to the ground
   (b) The required angle the boat needs to travel at to cross the river directly across
   (c) The time taken to cross the river
10. (3 points) A boat is crossing a river that is 10m wide with a velocity of $10\,\mathrm{ms}^{-2}$ relative to the water. However, the water is travelling at $4\,\mathrm{ms}^{-2}$ up relative to the ground. If the boat wants to travel in a straight line across, calculate:
    (a) The resultant velocity of the boat relative to the ground
    (b) The required angle the boat needs to travel at to cross the river directly across
    (c) The time taken to cross the river